Long-term livelihood impacts of floods: How measurement changes conclusions - Online Appendix

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A Additional Figures

Figure A1: GHSP data collection timeline

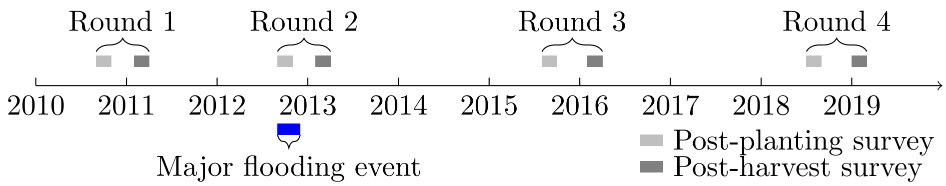
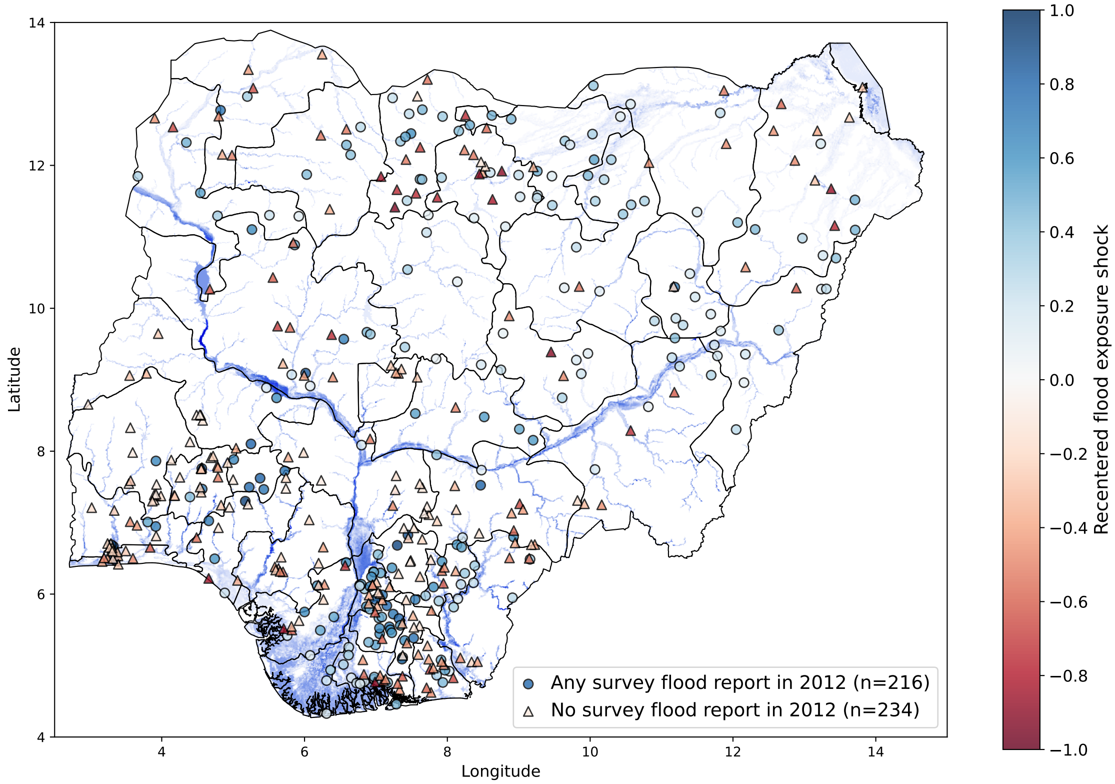
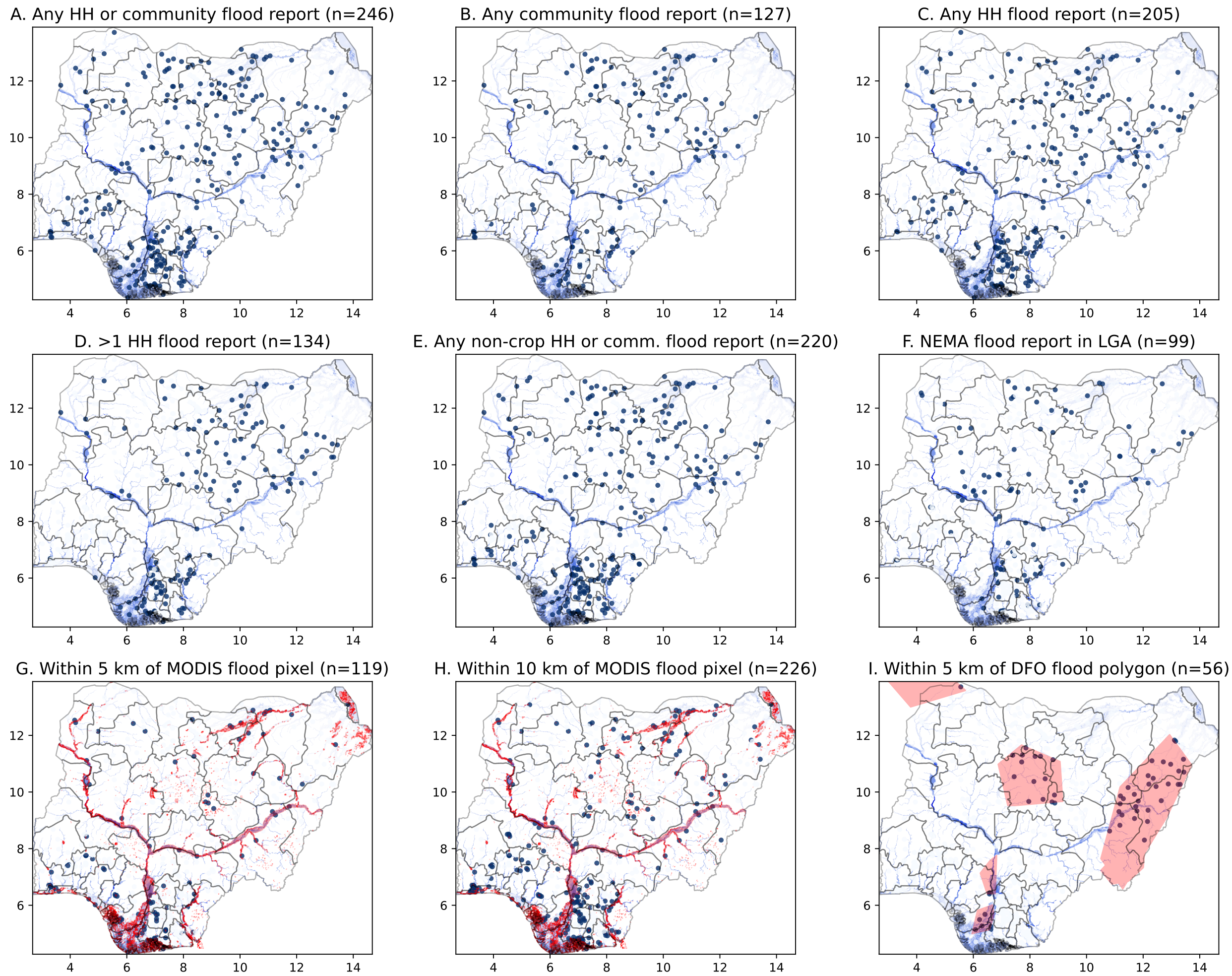


Figure A2: Recentered survey-reported 2012 community flood exposure, accounting for predicted 2012 exposure



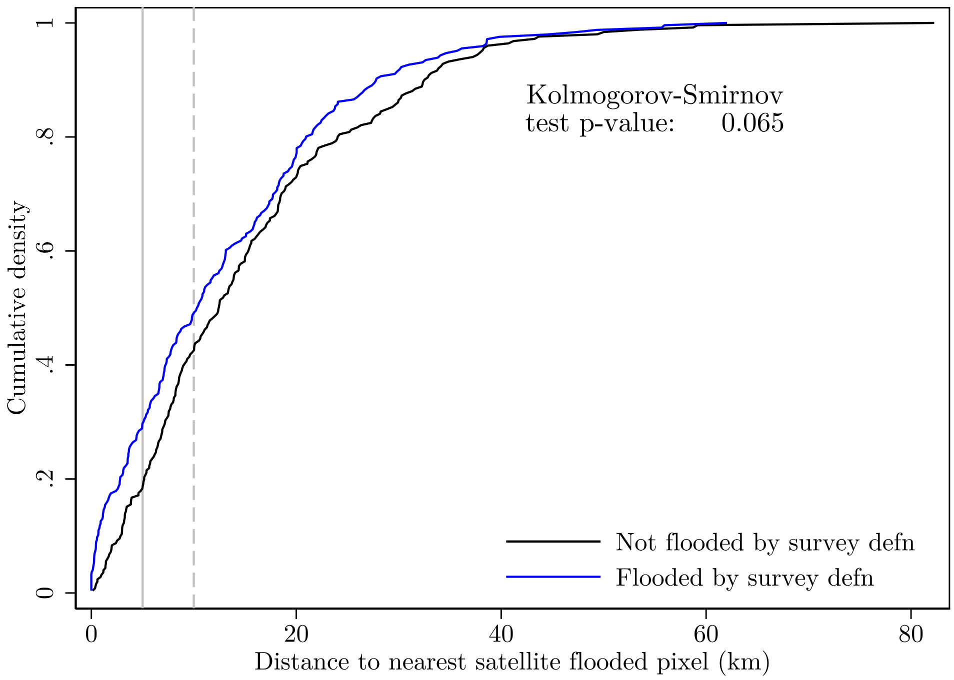
Note: The figure plots the locations of GHSP communities in the analysis sample by survey-reported 2012 community flood exposure and recentered flood treatment. The smaller number of flooded communities relative to Figure A3 Panel A is a result of balancing on recent flood history, underlying flood hazard, and predicted 2012 flood exposure. Flood exposure is defined as any household or community report of negative effects of floods within a community in 2012. The recentered flood treatment substracts the estimated 2012 flood propensity from the binary exposure. A value of 1 indicates a flooded community with an estimated flood propensity of 0, and a value of -1 indicates a non-flooded community with an estimated flood propensity of 1. The blue shading in the background corresponds to the depth of 100-year fluvial floods.

Figure A3: Community flood detection by flooding measure



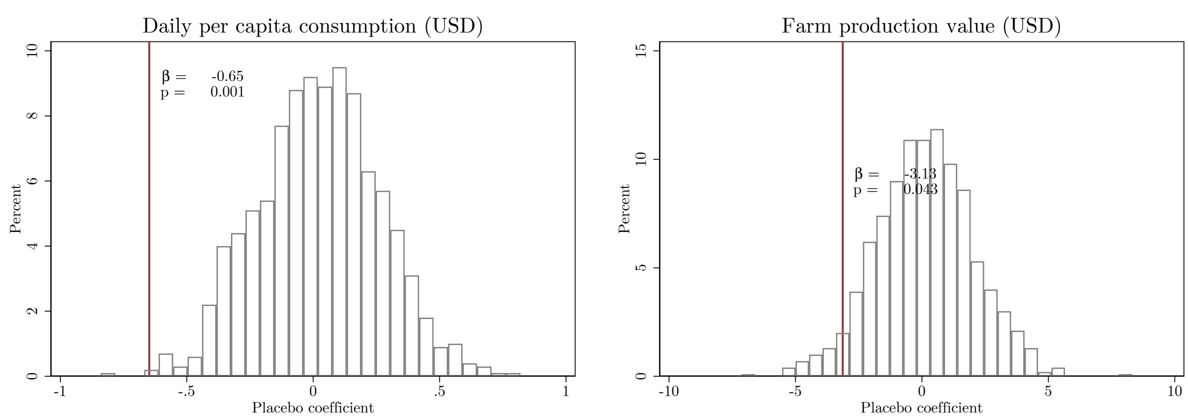
Note: Each panel shows the set of GHSP communities identified as exposed to floods in 2012 according to a different definition of flood exposure. The blue shading in the background of each panel corresponds to the depth of 100-year fluvial floods. Panels A-E rely on survey reports of floods in 2012. Panel A shows communities with any household or community survey flood report. Panel B shows communities with any community informant flood reports. Panel C shows communities with any household reports of floods in any module. Panel D shows communities with at least two such household reports. Panel E shows communities with any household non-crop-related flood or community flood report. Panel F shows communities in local government areas (LGAs) where NEMA reports any floods in 2012. Panels G and H show communities within 5 and 10 km of any pixel identified as flooded by MODIS satellite imagery at any point in 2012 in the NASA Global Flood Product. Panel I shows communities within 5 km of polygons where flooding is reported in the Dartmouth Flood Observatory archive. The flooded pixels and polygons in panels G-I are colored red.

Figure A4: Distance to nearest flooded pixel by survey flood definition



Note: The figure plots the cumulative distribution of GHSP community distance to the nearest pixel identified as flooded at any point in 2012 in the NASA MODIS flood database by whether the community was exposed to the 2012 floods according to our main survey-based definition. For each community, we calculate straight-line distances between the jittered community centroid and the centroids of 250 m MODIS flooded pixels. The solid vertical line represents the 5 km cutoff used in the main satellite-based flood exposure definition, and the dashed vertical line represents a 10 km cutoff. We also show the p-value for the Kolmogorov-Smirnov test of equality of the two distributions.

Figure A5: Distribution of impacts of permutated 2012 survey-reported flood exposure within flood propensity bins



Note: The two panels show the results of 1000 simulations of the impacts of 2012 recentered flood exposure permutating actual exposure within bins of estimated flood propensity to keep the same number of flooded communities in each bin. For each outcome, we plot the distribution of the estimated coefficients from each simulation. The vertical red line shows the coefficient for the effect of actual realized flood exposure. The text alongside the red line shows the coefficient estimate and the randomization p-value, calculated as the share of simulation estimates more extreme (in absolute terms) than the observed estimate. A small p-value indicates a very low likelihood that we would observe our estimate under the null of random allocation of flood exposure within flood propensity bins.

B Additional Tables

Table A1: Predictors of 2012 flood exposure

	(1) Any HH or comm flood rept	(2) Any HH flood rept	(3) >1 HH flood rept	(4) HH non-crop or comm. flood rept	(5) <5 km from MODIS flood	(6) <10 km from MODIS flood
Distance to nearest water area (km)	-0.012**	-0.015**	-0.023***	-0.003	-0.020**	-0.031***
	(0.005)	(0.006)	(0.007)	(0.004)	(0.009)	(0.006)
Avg 100-year flood depth within 5 km	0.508**	0.750**	0.623***	0.467***	2.579***	3.190***
	(0.216)	(0.315)	(0.192)	(0.165)	(0.649)	(0.802)
Area equipped for irrigation in cell in 2005 (100 ha)	0.008	0.009	0.012*	0.012**	0.005	0.009
	(0.007)	(0.007)	(0.007)	(0.006)	(0.007)	(0.008)
Percent ag land w/in approx 1 km	0.007	0.013*	0.003	0.007	-0.020	-0.017**
	(0.006)	(0.008)	(0.006)	(0.004)	(0.013)	(0.007)
Sparse vegetation (¡15%)	0.000	1.963	0.000	1.114	2.128	0.973
	(.)	(1.390)	(.)	(1.246)	(1.765)	(1.635)
>50% artificial surfaces and associated urban areas	-0.609	-0.619	-1.211		0.078	0.824
	(0.489)	(0.625)	(0.858)		(0.862)	(0.562)
Platform terrain (very low plateaus)	-0.692*	-0.320		-0.311		0.927**
	(0.395)	(0.499)		(0.356)		(0.384)
Low plateaus terrain		-1.883*	1.299*		-1.914	1.011
		(1.055)	(0.777)		(1.869)	(0.810)
Slope (percent)	-0.013	-0.020	-0.130*	-0.092*	-0.086	-0.064
	(0.061)	(0.098)	(0.073)	(0.053)	(0.125)	(0.070)
Total precipitation of wettest quarter in 2012 (100 mm)	0.162*	0.338	0.313**	0.217**		
	(0.084)	(0.333)	(0.127)	(0.108)		
Avg. precipitation of wettest quarter historically (mm)		-0.006**	0.002	0.001	-0.000	
		(0.003)	(0.002)	(0.001)	(0.002)	
Potential Wetness Index	-0.037	-0.020		-0.038	-0.032	
	(0.030)	(0.035)		(0.029)	(0.047)	
Percent ag land X 2012 deviation from avg. annual precipitation (100 mm)	0.292		0.204			0.165
	(0.200)		(0.212)			(0.410)
Percent ag land X 2012 deviation from avg. wettest qtr. precip. (100 mm)		0.004		0.004**	0.021***	0.011*
		(0.003)		(0.002)	(0.005)	(0.007)
Elevation (100 m) X 2012 deviation from avg. annual precipitation (100 mm)	3.998	15.915**	3.758		3.504	
	(3.650)	(6.586)	(2.629)		(6.886)	
HH distance to nearest market (km)	0.004	0.004			0.001	0.006*
	(0.003)	(0.004)			(0.005)	(0.004)
Any HH rept. flood shock in 2007		-0.862		-1.119**	-2.195*	-1.381**
		(0.614)		(0.548)	(1.265)	(0.682)
Any HH rept. flood shock in 2008	0.410	0.974	0.648	1.126**	-0.714	-0.900
	(0.611)	(0.696)	(0.635)	(0.555)	(1.063)	(0.704)
Any HH rept. flood shock in 2009	1.175***	1.149**	1.422***	0.943***	0.991*	
	(0.436)	(0.447)	(0.399)	(0.359)	(0.580)	
Any HH rept. flood shock in 2010	-0.682*	-0.327			0.899*	
	(0.361)	(0.396)			(0.525)	
Any HH rept. flood shock in 2011		0.441	0.099		0.228	1.224**
		(0.441)	(0.432)		(0.564)	(0.476)
Share of HHs w/ any crop activity in 2010	0.955**	1.998***	1.762***	0.588*	-0.763	
	(0.426)	(0.673)	(0.630)	(0.333)	(0.767)	
Mean HH area planted with crops in 2010 (ha)	0.055	0.341**	0.283**		0.028	-0.142
	(0.126)	(0.152)	(0.129)		(0.204)	(0.145)
Share of HHs experiencing any food insecurity in 2010		0.768	1.490**		-0.648	1.114*
		(0.662)	(0.680)		(0.881)	(0.637)
Observations	493	496	481	497	472	488

Note: This table shows the results of logit regressions predicting 2012 flood exposure under different measurement approaches indicated in the column headings. See Figure A3 for a description of the different measures. The predictors are selected from a broader set of geographic, weather, and mean household characteristics in a first stage using a Lasso regression with 10-fold cross-validation. Differences in the number of community observations result from certain selected land cover classes predicting failure or success perfectly, and observations in those land cover classes dropping out from the estimation. For simplicity, we only show characteristics selected for at least four flood measures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Estimated average impacts of 2012 flood exposure by flood measure

	Any HH or comm flood rept (1)	Any HH flood rept (2)	>1 HH flood rept (3)	HH non-crop or comm. flood rept (4)	<5 km from MODIS flood (5)	<10 km from MODIS flood (6)
Any food insecurity in last 12 months	-0.02	0.00	0.04	-0.01	0.02	-0.01
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)
Daily HH consumption per capita (USD)	-0.65***	-0.78***	-0.61**	-0.46*	0.20	-0.29
	(0.22)	(0.26)	(0.25)	(0.24)	(0.33)	(0.25)
Total value of HH farm production (USD 100s)	-3.13*	-2.62	-4.03**	-2.96	-3.78**	-1.53
	(1.74)	(1.85)	(1.75)	(2.36)	(1.92)	(1.60)
Total non-farm HH enterprise earnings (USD 100s)	-4.00	-2.88	2.94	-1.28	12.14***	0.92
	(3.01)	(3.50)	(3.44)	(3.01)	(4.63)	(3.92)
Total wage employment earnings (USD 100s)	4.69	-1.30	1.51	4.31	-1.05	9.08
	(5.39)	(5.53)	(6.16)	(5.97)	(8.86)	(6.44)
HH farm share of total HH earnings and prod value	-0.00	-0.01	-0.01	0.00	-0.02	-0.01
	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.02)
NFE share of total HH earnings and prod value	0.01	0.02	0.02	0.01	0.03	0.00
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Wage share of total HH earnings and prod value	-0.02	-0.02*	-0.00	-0.02	-0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)
HH earnings Herfindahl-Hirschman Index	0.00	-0.01	-0.00	0.01	0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Count of HH members wkg. in HH farm	0.08	0.05	0.07	0.08	-0.02	-0.09
	(0.09)	(0.09)	(0.11)	(0.08)	(0.11)	(0.08)
Count of HH members wkg. in HH non-farm enterprise	0.06	0.05	-0.02	0.05	0.04	0.05
	(0.06)	(0.05)	(0.06)	(0.06)	(0.05)	(0.05)
Count of HH members wkg. in wage employment	0.14***	0.07	0.10	0.12**	0.03	-0.12*
	(0.05)	(0.06)	(0.06)	(0.05)	(0.07)	(0.07)
Any member left HH for work during/since last round	-0.01	-0.00	-0.00	-0.01	-0.02	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. Each column indicates how 2012 community flood exposure is defined. For each flood measure, we estimate the 2012 flood propensity and use this to recenter flood exposure as described in **??**. Column (1) presents the results using the main survey definition of any 2012 flood report in any household survey module or in the community survey. Column (2) is similar but does not consider floods reported in the community survey, while column (3) further requires at least 2 household flood reports. Column (4) follows column (1) but excludes communities where floods are only reported in the household agriculture module or crop failure shock questions from the flood definition and from the sample. Columns (5) and (6) define communities as flooded based on proximity to pixels identified as flooded by MODIS satellite imagery from the NASA Global Flood Product, using 5 and 10 km buffers around the community centroid, respectively. All regressions include household and state-by-round fixed effects and the main analysis sample that is balanced on underlying flood hazard by 2007-2011 survey flood report strata. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Estimated average impacts of 2012 flood exposure by flood measure in the same overlapping sample

	Survey (1)	Satellite (2)
Daily HH consumption per capita (USD)	-0.63**	0.10
	(0.25)	(0.34)
Total value of HH farm production (USD 100s)	-2.16	-3.85**
	(1.82)	(1.92)
Total crop production value (USD 100s)	-3.11*	-4.22**
	(1.70)	(1.95)
Total non-farm HH enterprise earnings (USD 100s)	-3.23	10.72**
	(3.44)	(4.66)
Total wage employment earnings (USD 100s)	2.80	-0.02
	(5.83)	(9.40)
Count of HH members wkg. in HH farm	0.05	-0.02
	(0.10)	(0.11)
Count of HH members wkg. in HH non-farm enterprise	0.01	0.06
	(0.06)	(0.06)
Count of HH members wkg. in wage employment	0.12**	0.04
	(0.06)	(0.07)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. We consider the main survey and satellite measures of community flood exposure, and restrict the sample for both to the set of communities where both predicted 2012 flood propensities are defined. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. All regressions include household and state-by-round fixed effects and the main analysis sample that is balanced on underlying flood hazard by 2007-2011 survey flood report strata. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Balance in baseline characteristics by 2012 flood exposure

	No flood report Mean (SD)	Flood report difference (SE)	Recentered flood shock difference (SE)
Community characteristics			
Rural	0.62	0.10**	-0.10**
	(0.49)	(0.05)	(0.05)
HH distance to nearest major road (km)	13.22	1.46	1.94
	(17.95)	(2.21)	(1.66)
HH distance to nearest market (km)	66.72	6.64*	4.05
	(46.60)	(3.87)	(3.12)
HH distance to nearest pop. center w/ >20k pop. (km)	17.56	1.41	-3.28*
	(19.29)	(2.23)	(1.80)
Percent ag land w/in approx 1km	24.56	7.27***	1.25
	(26.10)	(2.73)	(2.18)
Slope (percent)	3.33	-0.21	-0.31*
	(2.67)	(0.27)	(0.18)
Elevation (m)	271.00	-24.52	-17.33
	(193.38)	(15.96)	(11.35)
Annual Precipitation (mm)	1431.95	18.82	-21.25
	(593.33)	(22.75)	(21.05)
Distance to nearest water area (km)	37.29	-3.19	-0.18
	(31.74)	(2.34)	(2.37)
Avg 100-year flood depth within 5 km	0.20	0.11*	-0.07
	(0.44)	(0.06)	(0.05)
Any comm. HH flood shock rept. from 2007-2011	0.19	0.10*	-0.04
	(0.39)	(0.05)	(0.04)
Household characteristics			
Female-headed HH	0.16	-0.01	-0.02
	(0.37)	(0.01)	(0.02)
Count of HH members	5.24	0.33**	0.06
	(2.95)	(0.16)	(0.13)
Household asset index	0.07	-0.22***	0.01
	(1.07)	(0.06)	(0.06)
Any food insecurity in last 12 months	0.20	0.02	0.02
	(0.40)	(0.02)	(0.02)
Daily HH consumption per capita (USD)	5.84	-0.41	0.28
	(5.01)	(0.26)	(0.24)
Household activities			
Any HH farm activity	0.64	0.10***	-0.06*
	(0.48)	(0.03)	(0.03)
Any HH non-farm enterprise activity	0.76	0.03	-0.00
	(0.43)	(0.02)	(0.02)
Any wage employment activity	0.42	-0.09***	0.01
	(0.49)	(0.03)	(0.02)
Total value of HH farm production (USD 100s)	15.36	6.88***	-1.40
	(36.52)	(1.71)	(1.98)
Total non-farm HH enterprise earnings (USD 100s)	26.54	-5.55*	3.28
	(83.20)	(3.02)	(2.97)
Total wage employment earnings (USD 100s)	34.37	-17.90***	-4.32
	(158.96)	(6.19)	(6.17)
Total area planted (ha)	1.23	0.92***	0.04
	(2.79)	(0.23)	(0.19)
Test of joint significance		$F = 7.69$	$F = 1.00$
p-value		$p < 0.001$	$p = 0.462$

Note: This table shows the baseline (2010-11) control mean and difference by 2012 flood exposure status for selected community and household characteristics from a series of separate regressions. All regressions include state fixed effects, and standard errors are clustered at the community level ($N = 497$). Flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. The first column of differences uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood exposure. The bottom of the table shows results for the test of the hypothesis that the relationship between a given 2012 flood exposure measure and all variables is jointly 0. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Balance in flood risk by 2012 flood exposure

	(1)	(2)
	Any flood report	
2012 reported flood exposure	0.015	
	(0.010)	
2012 recentered flood exposure		-0.000
		(0.011)
Observations	4205	4205
Mean, not flooded in 2012	0.042	0.042
State and Year FE	Yes	Yes

Note: This table shows the results from regressing a dummy for any community flood exposure in a given year according to the survey measure on dummies for flood exposure in 2012. The first row uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood exposure. Observations are at the community-year level for 2007-2018, and 2012 is omitted. Fixed effects control for state and year. Standard errors are clustered by community, but are similar if we cluster by year. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Separating effects of flood exposure and estimated flood risk

	Flood Shock (1)	Flood Propensity (2)
Any food insecurity in last 12 months	-0.02	0.00
	(0.02)	(0.06)
Daily HH consumption per capita (USD)	-0.65***	0.13
	(0.22)	(0.70)
Total value of HH farm production (USD 100s)	-3.13*	-0.52
	(1.74)	(6.68)
Total non-farm HH enterprise earnings (USD 100s)	-4.00	2.27
	(3.01)	(15.09)
Total wage employment earnings (USD 100s)	4.69	35.99
	(5.39)	(27.65)
HH farm share of total HH earnings and prod value	-0.00	-0.04
	(0.01)	(0.04)
NFE share of total HH earnings and prod value	0.01	-0.01
	(0.02)	(0.06)
Wage share of total HH earnings and prod value	-0.02	0.07*
	(0.01)	(0.05)
HH earnings Herfindahl-Hirschman Index	0.00	-0.02
	(0.01)	(0.04)
Count of HH members wkg. in HH farm	0.08	0.03
	(0.09)	(0.27)
Count of HH members wkg. in HH non-farm enterprise	0.06	0.11
	(0.06)	(0.23)
Count of HH members wkg. in wage employment	0.14***	-0.70**
	(0.05)	(0.29)
Any member left HH for work during/since last round	-0.01	0.11***
	(0.01)	(0.04)

Note: This table compares estimated effects of 2012 flood exposure following the main specification (column 1) to effects of 2012 flood propensity among non-flooded communities (column 2). The purpose of the comparison is to determine to what extent flood propensity alone may be associated with different trends over time after the major 2012 floods. In addition to dropping flooded communities from the sample, the specification in column 2 follows the main specification but replaces the recentered 2012 flood shock × post term with a 2012 estimated flood propensity × post term. Since the recentering of the flood shock in column 1 directly accounts for this flood propensity, the estimated effects control for any differential trends post-2012 by flood propensity, isolating impacts of actual flood exposure. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Sensitivity to alternative ways of controlling for non-random component of survey-reported 2012 community flood exposure

	(1)	(2)	(3)	(4)	(5)
	Recentering	No recentering			
	Main (SE)	Prediction-round controls (SE)	Risk/rain-round controls (SE)	No added controls (SE)	No added controls (SE)
Any food insecurity in last 12 months	-0.02	-0.02	-0.02	-0.02	-0.00
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Daily HH consumption per capita (USD)	-0.65***	-0.69***	-0.75***	-0.74***	-0.67***
	(0.22)	(0.22)	(0.22)	(0.22)	(0.19)
Total value of HH farm production (USD 100s)	-3.13*	-3.17*	-3.04*	-2.96*	-3.29**
	(1.74)	(1.74)	(1.61)	(1.62)	(1.58)
Total non-farm HH enterprise earnings (USD 100s)	-4.00	-3.87	-2.92	-3.09	-2.55
	(3.01)	(2.98)	(2.86)	(2.83)	(2.64)
Total wage employment earnings (USD 100s)	4.69	5.66	6.79	7.71	7.54
	(5.39)	(5.34)	(5.00)	(5.08)	(4.65)
HH farm share of total HH earnings and prod value	-0.00	-0.00	-0.01	-0.01	-0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
NFE share of total HH earnings and prod value	0.01	0.01	0.01	0.01	0.01
	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)
Wage share of total HH earnings and prod value	-0.02	-0.02	-0.01	-0.01	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
HH earnings Herfindahl-Hirschman Index	0.00	0.00	0.00	0.00	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Count of HH members wkg. in HH farm	0.08	0.08	0.07	0.07	0.10
	(0.09)	(0.09)	(0.09)	(0.09)	(0.08)
Count of HH members wkg. in HH non-farm enterprise	0.06	0.06	0.05	0.05	0.01
	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)
Count of HH members wkg. in wage employment	0.14***	0.13**	0.07	0.07	0.09*
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Any member left HH for work during/since last round	-0.01	-0.01	-0.00	-0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Sample	Main	Main	Main	Main	All EAs

Note: This table repeats Table A2 column 1 but varies how we control for the non-random component of survey-reported 2012 community flood exposure. Each column indicates how the specification is changed relative to the main specification, presented in column (1), where the 2012 survey flood exposure measure is recentered around its predicted value. Columns (2)-(5) show effects of the binary 2012 survey flood measure with no recentering and varying controls and sample restrictions. Columns (2)-(4) keep the main sample restrictions: communities within the common support of average flood risk stratified by the number of years of survey-reported flooding from 2007-2011, and within the common support of the predicted 2012 flood exposure measure by actual flood exposure. Column (2) adds predicted 2012 survey flood exposure-by-round controls. Column (3) adds average 100-year flood depth within 5 km-by-round and community 2012 rainfall deviation-by-round controls. Column (4) includes no additional controls. Column (5) also includes no additional controls and expands the sample to include all communities (EAs), with no exclusion based on common support of average flood risk or predicted 2012 flooding. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to alternative specifications

	Main (1)	Add unbalanced controls (2)	Add rain controls (3)	Round FE (4)
Any food insecurity in last 12	-0.02	-0.02	-0.02	0.02
months	(0.02)	(0.02)	(0.02)	(0.03)
Daily HH consumption per capita	-0.65***	-1.00***	-0.63***	-0.56**
(USD)	(0.22)	(0.32)	(0.21)	(0.24)
Total value of HH farm production	-3.13*	-3.25*	-3.21*	-3.38*
(USD 100s)	(1.74)	(1.97)	(1.74)	(1.87)
Total non-farm HH enterprise	-4.00	-4.41	-4.01	-3.59
earnings (USD 100s)	(3.01)	(3.52)	(2.99)	(3.10)
Total wage employment earnings	4.69	4.56	4.77	3.49
(USD 100s)	(5.39)	(8.38)	(5.37)	(5.48)
HH farm share of total HH earnings	-0.00	-0.03	-0.00	-0.01
and prod value	(0.01)	(0.02)	(0.01)	(0.02)
NFE share of total HH earnings and	0.01	0.02	0.01	0.03
prod value	(0.02)	(0.02)	(0.02)	(0.02)
Wage share of total HH earnings	-0.02	-0.01	-0.02	-0.02*
and prod value	(0.01)	(0.01)	(0.01)	(0.01)
HH earnings Herfindahl-Hirschman	0.00	-0.00	0.00	0.01
Index	(0.01)	(0.01)	(0.01)	(0.01)
Count of HH members wkg. in HH	0.08	0.02	0.08	0.09
farm	(0.09)	(0.11)	(0.09)	(0.09)
Count of HH members wkg. in HH	0.06	0.00	0.06	0.07
non-farm enterprise	(0.06)	(0.07)	(0.06)	(0.05)
Count of HH members wkg. in wage	0.14***	0.17**	0.14***	0.16**
employment	(0.05)	(0.07)	(0.05)	(0.07)
Any member left HH for work	-0.01	0.00	-0.01	-0.01
during/since last round	(0.01)	(0.01)	(0.01)	(0.02)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. The columns show average effects of flood exposure across all post-flood survey rounds. Flood exposure is measured using the main survey definition of any 2012 flood report in any household survey module or in the community survey, and is recentered around predicted survey-reported 2012 community flood exposure. Each column indicates how the specification is changed relative to the main specification, presented in column (1), which includes household and state-by-round fixed effects but no controls. In column (3) we add controls for baseline characteristics that differ significantly by recentered flood status in Table A4, interacted with survey round. In column (3) we instea add controls for the deviation in annual total precipitation relative to the historical mean at the community location. Columns (4) replace the state-by-round FE with round FE. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to alternative samples

	Main	All EAs	W/in 50 km of any 2012 flood report	Main, drop 20km donut	Main, drop LGA donut
	(1)	(2)	(3)	(4)	(5)
Any food insecurity in last 12 months	-0.02	-0.01	-0.01	-0.03	-0.03
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Daily HH consumption per capita (USD)	-0.65***	-0.63***	-0.61***	-0.61***	-0.49**
	(0.22)	(0.21)	(0.21)	(0.22)	(0.22)
Total value of HH farm production (USD 100s)	-3.13*	-3.45**	-3.35*	-4.03**	-5.22***
	(1.74)	(1.76)	(1.77)	(1.88)	(1.60)
Total non-farm HH enterprise earnings (USD 100s)	-4.00	-4.10	-3.91	-2.87	-3.74
	(3.01)	(2.88)	(2.92)	(3.15)	(2.94)
Total wage employment earnings (USD 100s)	4.69	4.51	4.67	5.48	5.54
	(5.39)	(5.40)	(5.47)	(4.94)	(5.37)
HH farm share of total HH earnings and prod value	-0.00	-0.01	-0.01	-0.01	-0.00
	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)
NFE share of total HH earnings and prod value	0.01	0.02	0.02	0.02	0.02
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Wage share of total HH earnings and prod value	-0.02	-0.02*	-0.02*	-0.02	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
HH earnings Herfindahl-Hirschman Index	0.00	0.00	0.00	0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Count of HH members wkg. in HH farm	0.08	0.10	0.09	0.06	0.11
	(0.09)	(0.08)	(0.08)	(0.09)	(0.09)
Count of HH members wkg. in HH non-farm enterprise	0.06	0.03	0.03	0.09	0.08
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Count of HH members wkg. in wage employment	0.14***	0.15***	0.16***	0.15***	0.11**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)
Any member left HH for work during/since last round	-0.01	-0.01	-0.01	-0.02	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Observations	11691	12328	12039	10770	10088

Note: This table repeats Table A8 but varies the sample instead of the specifications. Each column indicates how the sample is changed relative to the main specification, presented in column (1), which restricts the sample to communities within the common support of average 100-year flood depth within 5 km after stratifying by the number of years with any survey flood report from 2007-2011. In column (2) the community (EA) sample restrictions are removed. Column (3) replaces the main sample restriction with dropping any community more than 50 km from any community where floods were reported in 2012. Columns (4) and (5) add sample restrictions dropping communities which may be most affected by flood exposure spillovers: those less than 20 km from any community where a flood was reported in 2012, and those in the same LGA as a flooded sample community. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to different standard error clustering

	(1) Daily HH cons./cap.	(2) HH farm prod. val.	(3) HH crop prod. val.	(4) HH NFE earnings	(5) HH wage earnings	(6) Ct. wage wkrs.
Flood Exposure X Post	-0.648***	-3.131*	-4.497***	-4.001	4.693	0.144***
	(0.217)	(1.743)	(1.499)	(3.012)	(5.387)	(0.054)
SE: LGA	0.218	1.705	1.435	2.937	5.158	0.055
SE: Comm-Round	0.212	1.421	1.027	2.871	-	-
SE: Conley (50km)	0.219	1.366	1.186	2.537	3.841	0.040
SE: Conley (100km)	0.209	1.352	1.208	2.517	3.145	0.040
Wild p-val: State	0.038	0.090	0.006	0.195	0.436	0.021
Observations	11691	11691	11691	11691	11691	11691

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. Each regression follows the main specification, but we show how the standard error changes under different clustering approaches. The main regressions cluster SEs at the level of the community of residence in 2012, at which the flood exposure treatment is assigned. We then show SEs clustered at the LGA level, two-way clustered SEs at the community by round level, Conley SEs allowing for spatial correlation within 50 and 100 km, and p-values from a wild bootstrap of state-level clustered SEs. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Differences in coping strategies by flood exposure among households reporting shocks

	N	Non-flooded Mean (SD)	Flood shock (SE)
Sale of livestock	1888	0.22	0.15***
		[0.42]	(0.04)
Sale of land/property	1888	0.16	0.09**
		[0.36]	(0.03)
Sent kids away	1888	0.05	0.03
		[0.22]	(0.02)
Withdrew kids from school	1888	0.04	0.01
		[0.19]	(0.02)
Increased enterprise activity	1888	0.05	0.04*
		[0.22]	(0.02)
Assistance from friends/family	1888	0.28	0.05
		[0.45]	(0.04)
Borrowed from friends/family	1888	0.23	0.03
		[0.42]	(0.04)
HH member migrated for work	1888	0.03	0.02
		[0.17]	(0.02)
Took loan or credit	1888	0.18	-0.00
		[0.39]	(0.03)
Reduced consumption	1888	0.22	0.06*
		[0.41]	(0.03)
Received other assistance	1888	0.01	0.03**
		[0.12]	(0.02)

Note: This table shows the results from separate regressions of indicators that households reporting any shock in the 2013 post-harvest survey declare different coping strategies on an indicator for whether they declare any flood shock, controlling for whether they report other types of shocks. All regressions include community fixed effects. Standard errors are clustered by community. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Impacts of survey-reported 2012 community flood exposure on household consumption and assets

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Daily HH consumption per adult	11676	6.52	-0.61***	-0.62*	-0.60**	-0.63*
equiv (USD)		[4.91]	(0.23)	(0.36)	(0.25)	(0.34)
Daily HH consumption per capita	11691	5.64	-0.65***	-0.83***	-0.47*	-0.46
(USD)		[4.40]	(0.22)	(0.30)	(0.25)	(0.30)
Daily HH food cons. per capita	11691	4.35	-0.64***	-0.69***	-0.58***	-0.64***
(USD)		[4.43]	(0.18)	(0.21)	(0.22)	(0.24)
Daily HH non-food cons. per capita	11691	1.29	-0.01	-0.14	0.11	0.19
(USD)		[2.79]	(0.18)	(0.27)	(0.20)	(0.28)
HH under 1.90 USD PPP per capita	11691	0.32	0.05**	0.05**	0.04	0.08**
daily poverty line		[0.47]	(0.02)	(0.02)	(0.03)	(0.03)
Value of HH assets (USD)	11691	2356.56	-86.71	-94.69	-94.96	5.45
		[5520.11]	(152.13)	(173.28)	(176.48)	(367.08)
Livestock holdings (TLU)	11691	2.72	1.57*	1.41	1.74**	1.66**
		[42.95]	(0.87)	(0.94)	(0.86)	(0.78)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A13: Impacts of survey-reported 2012 community flood exposure on household earnings and production value by activity

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Total HH earnings and production	11691	89.42	-0.99	3.45	-5.16	-5.43
value (USD 100s)		[205.00]	(6.50)	(7.62)	(6.70)	(11.94)
Total value of HH farm production	11691	18.12	-3.13*	-4.02**	-2.17	-2.92
(USD 100s)		[40.83]	(1.74)	(1.88)	(2.00)	(2.72)
Total crop production value (USD	11691	15.18	-4.50***	-4.68***	-4.40**	-3.91
100s)		[32.69]	(1.50)	(1.63)	(1.81)	(2.55)
Total livestock production value	11691	2.93	1.37*	0.66	2.23**	0.98
(USD 100s)		[23.19]	(0.82)	(0.82)	(0.87)	(1.49)
Total earnings from non-farm	11691	71.31	2.14	7.48	-2.99	-2.51
activities (USD 100s)		[204.28]	(6.38)	(7.41)	(6.63)	(12.06)
Total non-farm HH enterprise	11691	26.83	-4.00	-1.81	-6.97**	-1.32
earnings (USD 100s)		[77.46]	(3.01)	(3.26)	(3.44)	(5.17)
Total wage employment earnings	11691	40.63	4.69	7.48	2.25	1.02
(USD 100s)		[188.67]	(5.39)	(6.40)	(5.45)	(10.08)
Total HH earnings from other	11691	3.85	1.45*	1.81**	1.74	-2.21
activities (USD 100s)		[22.79]	(0.84)	(0.89)	(1.07)	(1.79)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A14: Impacts of survey-reported 2012 community flood exposure on household migration and roster changes

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Any member left HH during/since	11691	0.09	-0.01	-0.02	0.02	-0.08**
last round		[0.28]	(0.02)	(0.02)	(0.02)	(0.03)
Any member left HH for work	11691	0.02	-0.01	-0.01	-0.03**	0.06*
during/since last round		[0.13]	(0.01)	(0.01)	(0.02)	(0.03)
Any member left HH for family	11691	0.03	-0.03**	-0.04**	-0.03	-0.05
during/since last round		[0.16]	(0.02)	(0.02)	(0.02)	(0.04)
Any adult member joined HH	11691	0.08	0.02	0.02	0.02*	-0.00
during/since last round		[0.27]	(0.01)	(0.01)	(0.01)	(0.02)
Count of HH members	11691	5.36	0.03	0.02	0.03	0.11
		[2.83]	(0.06)	(0.05)	(0.07)	(0.20)
Count of HH members ages 0-4	11691	0.80	-0.04	-0.04	-0.04	-0.06
		[1.04]	(0.04)	(0.04)	(0.05)	(0.09)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. See Appendix C for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Impacts of survey-reported 2012 community flood exposure on wages and market food prices

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
HH avg. PP daily wage earned by wage wkrs (USD)	1964	27.82	-6.75**	-4.94	-8.45**	-12.06*
		[30.64]	(2.91)	(3.48)	(3.43)	(6.46)
HH avg. PH daily wage earned by wage wkrs (USD)	2072	32.78	6.24	9.57*	3.85	-3.48
		[43.35]	(4.19)	(5.42)	(4.99)	(5.23)
HH avg. daily wage paid to hired ag labor (USD)	4417	12.92	2.24	1.87	2.58*	2.45
		[16.89]	(1.57)	(1.78)	(1.50)	(1.61)
Comm. avg. daily wage for men's ag labor (USD)	9224	17.41	-0.48	-0.75	-0.31	0.35
		[9.20]	(0.92)	(1.20)	(0.94)	(1.74)
Comm. price shelled maize (USD/kg z-score)	8259	-0.10	0.10	0.25	-0.08	
		[0.63]	(0.14)	(0.17)	(0.12)	
Comm. price local rice (USD/kg z-score)	8530	-0.10	-0.06	0.15	-0.26**	
		[0.54]	(0.10)	(0.14)	(0.13)	
Comm. price bread (USD/kg z-score)	9933	-0.12	-0.32***	-0.29**	-0.35***	
		[0.65]	(0.11)	(0.13)	(0.13)	
Comm. price yam roots (USD/kg z-score)	9016	0.00	-0.07	0.01	-0.14	
		[0.95]	(0.12)	(0.13)	(0.14)	
Comm. price palm oil (USD/litre z-score)	9924	0.04	0.00	-0.08	0.09	
		[1.15]	(0.10)	(0.12)	(0.12)	
Comm. price groundnut oil (USD/litre z-score)	9475	0.07	0.12	0.13	0.11	
		[1.35]	(0.10)	(0.11)	(0.12)	
Comm. price banana (USD/kg z-score)	7625	0.03	0.11	-0.07	0.30	
		[0.96]	(0.16)	(0.16)	(0.20)	
Comm. price chicken (USD/kg z-score)	8313	0.06	0.05	-0.16	0.32**	
		[1.02]	(0.11)	(0.11)	(0.16)	
Comm. price sugar (USD/kg z-score)	8281	-0.06	0.11	0.09	0.12	
		[0.85]	(0.13)	(0.15)	(0.13)	

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. Household wages received or paid are only reported by households active in wage labor markets. Market food prices were not collected in the same way in 2018-19, so estimates for these outcomes exclude that round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A16: Impacts of survey-reported 2012 community flood exposure on average household physical yield and sales prices by crop

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Yield by area planted of cereals	4151	2181.91	-1082.85**	-1454.12**	-685.44	-583.84
(kgs/ha)		[4310.20]	(512.46)	(592.10)	(578.58)	(823.92)
Yield by area planted of maize	3048	1419.84	-613.29	-1343.98**	-24.08	-792.79*
(kgs/ha)		[3525.54]	(411.88)	(528.61)	(488.58)	(475.45)
Yield by area planted of sorghum	2869	1685.30	-236.93	-287.28	-178.02	-239.79
(kgs/ha)		[3246.21]	(422.54)	(487.21)	(558.96)	(615.93)
Yield by area planted of millet	1792	1170.63	-828.54	-917.29	-795.51	-105.53
(kgs/ha)		[1605.19]	(557.41)	(678.87)	(716.08)	(602.26)
Yield by area planted of rice	475	2714.51	-1583.35*	-1131.83	-1583.95	-5600.71***
(kgs/ha)		[2601.40]	(911.22)	(1313.15)	(971.16)	(1979.35)
Yield by area planted of cowpea	2000	593.18	-483.13**	-478.10*	-502.53*	-296.93
(kgs/ha)		[1196.41]	(211.35)	(249.49)	(295.09)	(391.35)
Yield by area planted of groundnut	858	1268.76	-1005.89**	-923.31	-1187.54	-577.87
(kgs/ha)		[2587.19]	(492.16)	(599.89)	(926.94)	(1604.91)
Yield by area planted of yam	2421	8899.53	-1692.69	-4471.32	704.08	4121.74
(kgs/ha)		[18059.75]	(2514.69)	(3703.42)	(3096.73)	(3244.46)
Yield by area planted of cassava	3254	1800.41	764.96	1094.43	219.97	1819.89
(kgs/ha)		[4145.43]	(832.23)	(1159.13)	(951.95)	(1228.69)
Average cereal sales price (USD/kg)	4141	0.55	-0.10	-0.08	-0.12	-0.06
		[1.12]	(0.08)	(0.07)	(0.12)	(0.20)
Average maize sales price (USD/kg)	2544	0.31	0.11	0.10	0.11	0.23
		[2.51]	(0.14)	(0.16)	(0.14)	(0.17)
Average sorghum sales price (USD/kg)	2814	0.99	0.02	0.06	-0.05	0.30**
		[0.93]	(0.09)	(0.06)	(0.15)	(0.15)
Average millet sales price (USD/kg)	1766	0.14	-0.19	-0.02	-0.37	-0.18
		[0.55]	(0.13)	(0.07)	(0.23)	(0.16)
Average rice sales price (USD/kg)	412	0.96	-0.21**	-0.22	-0.21	-0.17
		[0.54]	(0.09)	(0.20)	(0.20)	(0.19)
Average cowpea sales price (USD/kg)	1902	1.72	0.04	0.00	0.06	0.49
		[1.01]	(0.28)	(0.34)	(0.21)	(0.35)
Average groundnut sales price (USD/kg)	801	0.02	0.07	0.07	0.10	-0.09
		[0.22]	(0.07)	(0.10)	(0.09)	(0.12)
Average yam sales price (USD/kg)	2277	0.66	-0.03	-0.00	-0.03	-0.21
		[0.50]	(0.02)	(0.05)	(0.06)	(0.17)
Average cassava sales price (USD/kg)	976	0.68	0.33	0.25	0.42	0.46
		[1.20]	(0.27)	(0.29)	(0.31)	(0.29)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. We show results for the eight most commonly-cultivated crops in Nigeria across the full sample. Sample sizes vary based on the number of households cultivating each crop. Sample sizes for prices are smaller than for yields because of households not harvesting any of the crop they planted. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A17: Heterogeneity in impacts of survey-reported 2012 community flood exposure by baseline household and community characteristics

		HH agriculture		Wage work		HH NFE	
	(1) Daily HH cons./cap.	(2) Value prod.	(3) Count mems.	(4) Total earn.	(5) Count mems.	(6) Total earn.	(7) Count mems.
Flood × Post ×	0.18	-5.29	-0.06	-17.52*	0.22**	8.64	0.08
> 5 baseline HH members	(0.38)	(3.33)	(0.16)	(9.96)	(0.10)	(5.91)	(0.10)
Flood × Post ×	-0.10	-1.14	-0.13	15.73	-0.12	-0.68	-0.12
Above med. baseline HH assets value	(0.37)	(3.52)	(0.13)	(10.87)	(0.10)	(6.07)	(0.10)
Flood × Post ×	-0.32	-1.57	0.18	11.11	0.31***	14.98*	0.24**
Rural community	(0.49)	(2.72)	(0.15)	(16.87)	(0.10)	(7.72)	(0.10)
Flood × Post ×	-0.65	3.86	-0.01	0.79	-0.04	-15.11**	-0.18
Below med. dist. to major market	(0.48)	(3.49)	(0.19)	(10.63)	(0.12)	(6.47)	(0.12)
Flood × Post ×	0.07	-1.04	-0.22	23.56**	-0.02	-9.77*	-0.11
Below med. dist. to major road	(0.41)	(3.14)	(0.16)	(10.78)	(0.10)	(5.74)	(0.11)
Flood × Post ×	0.49	2.12	-0.12	3.74	-0.11	-11.30*	-0.06
Below med. dist. to > 20k pop. town	(0.42)	(3.22)	(0.17)	(10.25)	(0.10)	(5.83)	(0.12)
Flood × Post ×	-0.49	-6.55*	0.21	-1.17	-0.03	0.14	-0.05
Below med. dist. to river/water	(0.46)	(3.62)	(0.17)	(11.29)	(0.11)	(5.98)	(0.12)
Flood × Post ×	-0.22	-1.83	-0.21	-2.35	0.02	-2.28	-0.24*
Non-zero fluvial flood risk	(0.58)	(3.30)	(0.18)	(13.88)	(0.11)	(6.51)	(0.12)
Flood × Post ×	0.43	-2.82	-0.41	0.65	-0.04	12.27**	-0.24
Comm. flooded in last 5 years	(0.58)	(4.03)	(0.27)	(9.97)	(0.16)	(5.31)	(0.15)

Note: This table presents tests of heterogeneity in average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds by baseline household and community characteristics. Each column represents an outcome. Monetary values are in 2016 PPP USD. See Appendix C for details on variable construction. The results for each column in each row are from separate triple-differences regressions fully interacting the indicated baseline household or community characteristic with the Post and Flood dummies and state-by-round fixed effects. We show only the estimated coefficient for the interaction term for simplicity, reflecting the estimated difference in effects for households/communities with a given characteristic. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A18: Balance in baseline characteristics by baseline livelihood activities

	Non-ag Mean (SD)	Ag HH difference (SE)	Ag HH, Non-NFE Mean (SD)	NFE HH difference (SE)	Ag HH, Non-wage Mean (SD)	Wage HH difference (SE)
Community characteristics						
Rural	0.26	0.50***	0.89	-0.04***	0.87	-0.09***
	(0.44)	(0.03)	(0.32)	(0.02)	(0.33)	(0.02)
HH distance to nearest major road (km)	7.76	5.05***	18.39	-2.66***	18.29	-2.93***
	(14.31)	(1.25)	(19.80)	(0.92)	(20.07)	(0.87)
HH distance to nearest market (km)	55.58	10.90***	70.47	-3.02*	72.30	-2.97*
	(48.72)	(2.88)	(40.07)	(1.80)	(40.57)	(1.55)
HH distance to nearest pop. center w/ ¿20k pop. (km)	9.57	9.48***	24.85	-2.63***	24.82	-3.02***
	(12.78)	(1.24)	(21.13)	(0.87)	(21.33)	(0.87)
Percent ag land w/in approx 1km	14.61	10.33***	29.52	1.37	35.02	-0.74
	(19.88)	(1.74)	(24.95)	(0.89)	(28.55)	(0.95)
Slope (percent)	3.11	0.28	3.21	-0.01	3.24	-0.14
	(2.61)	(0.21)	(2.35)	(0.10)	(2.78)	(0.12)
Elevation (m)	216.69	9.14	274.27	4.14	305.27	12.26*
	(200.01)	(8.35)	(199.68)	(5.15)	(199.81)	(6.32)
Annual Precipitation (mm)	1608.39	-38.80***	1421.39	9.91	1334.49	-0.75
	(627.51)	(14.02)	(612.53)	(8.57)	(593.59)	(7.24)
Distance to nearest water area (km)	32.44	7.57***	33.41	0.00	33.61	1.24
	(31.40)	(1.94)	(28.28)	(1.25)	(29.39)	(1.02)
Avg 100-year flood depth within 5 km	0.38	-0.10**	0.17	-0.02	0.17	-0.01
	(0.65)	(0.04)	(0.42)	(0.02)	(0.39)	(0.01)
Any comm. HH flood shock rept. from 2007-2011	0.09	0.10***	0.24	0.03	0.30	0.01
	(0.29)	(0.03)	(0.43)	(0.02)	(0.46)	(0.02)
Household characteristics						
Female-headed HH	0.20	-0.06***	0.17	-0.05***	0.14	-0.06***
	(0.40)	(0.02)	(0.38)	(0.01)	(0.34)	(0.01)
Count of HH members	4.62	0.89***	5.21	0.94***	5.68	0.95***
	(2.49)	(0.13)	(3.08)	(0.13)	(3.01)	(0.12)
Household asset index	0.66	-0.65***	-0.29	0.11**	-0.34	0.36***
	(1.22)	(0.06)	(0.95)	(0.04)	(0.67)	(0.04)
Any food insecurity in last 12 months	0.20	-0.01	0.21	-0.03	0.18	-0.01
	(0.40)	(0.02)	(0.41)	(0.02)	(0.38)	(0.01)
Daily HH consumption per capita (USD)	7.97	-2.63***	4.90	-0.34*	4.42	0.28**
	(6.24)	(0.26)	(4.42)	(0.18)	(3.48)	(0.14)
Household activities						
Any HH non-farm enterprise activity	0.81	-0.02	0.00	1.00***	0.76	0.04***
	(0.39)	(0.02)	(0.00)	(0.00)	(0.43)	(0.02)
Any wage employment activity	0.53	-0.16***	0.33	0.06***	0.00	1.00***
	(0.50)	(0.02)	(0.47)	(0.02)	(0.00)	(0.00)
Total value of HH farm production (USD 100s)	0.00	26.21***	23.27	2.69	25.39	-0.74
	(0.00)	(1.94)	(47.52)	(2.05)	(38.95)	(1.61)
Total non-farm HH enterprise earnings (USD 100s)	52.31	-30.09***	0.00	22.09***	17.77	3.19
	(121.67)	(4.86)	(0.00)	(1.61)	(58.13)	(2.45)
Total wage employment earnings (USD 100s)	79.26	-55.43***	16.92	3.93	0.00	50.77***
	(234.13)	(10.52)	(54.75)	(3.71)	(0.00)	(5.40)
Total area planted (ha)	0.00	1.97***	1.94	0.17	2.39	0.03
	(0.00)	(0.15)	(3.42)	(0.15)	(3.83)	(0.15)

Note: This table shows the baseline (2010-11) control mean and difference by baseline livelihood activities for selected community and household characteristics from a series of separate regressions for the main analysis sample. All regressions include state fixed effects, and standard errors are clustered at the community level. We show three sets of balance tests. The first two columns show differences by baseline engagement in agriculture across all sample columns. The next two columns show differences by baseline engagement in NFE among households engaged in agriculture at baseline. The last two columns show differences by baseline engagement in wage work among households engaged in agriculture at baseline. In all three sets of columns, the first column shows the mean for the reference group and the second column shows the coefficient on the particular baseline activity from a regression of that activity on the outcome at in 2010-11. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C Survey Data

The main data source for the analysis is Nigeria's General Household Survey Panel (GHSP). The GHSP, part of the World Bank's Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA), is a nationally-representative panel survey including 5,000 households conducted by the Nigeria National Bureau of Statistics. Before the first survey round, 500 enumeration areas (communities) were randomly sampled after stratifying by state. Ten households were then randomly sampled in each enumeration area from a census of households.

Four survey rounds were conducted between 2010 and 2019. Households are tracked over time, including if they move to a new location, but individual household members are not tracked. In 2018-19 the panel sample was partially refreshed, with 1,425 households from the original panel retained (clustered in randomly selected panel communities) and 3,551 new panel households added to the sample. The GHSP data are publicly available from the World Bank's Microdata Catalog.

Each survey round includes both a post-planting and a post-harvest survey. Post-planting surveys took place in September-October 2010, September-October 2012, August-September 2015, and July-August 2018. Post-harvest surveys took place in February-April of 2011, 2013, and 2016 and in January-February 2019. Some information, such as individual labor supply, is recorded in each survey while other modules are only included in either the post-planting or post-harvest surveys.

Basic cleaning decisions include replacing missing values with 0 where appropriate (i.e., for income in an activity the household was not engaged in) and replacing impossible values (such as more than 24 hours per day) with missing values. We winsorize continuous variables by replacing values above the 99th percentile with the value at the 99th percentile. All variables representing monetary values are converted to 2016 PPP USD.

For certain variables – notably the measures of total household earnings by activity – we modify code developed by the Evans School Policy Analysis & Research Group (EPAR) to construct variables in consistent ways accounting for differences in the survey instruments across rounds. The code is available on GitHub: https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Agricultural-Indicators-Code.

Construction of outcome variables

We report impacts on a wide variety of outcomes. Daily household consumption outcomes are calculated on the basis of reported consumption across different goods and services, and are provided directly in the publicly-available GHSP data. Daily consumption is divided by either the count of household members or the count of adult equivalents. Poverty is calculated based on per capita daily consumption. The value of household assets is the sum of declared values across all assets owned. Livestock are measured in tropical livestock units (TLU) in order to aggregate across species.

Changes in household composition are measured in reference to the previous survey round, as the household roster is updated in each survey and respondents can indicate which members have left and their reason for leaving and if any new members have joined. Counts of household members working in different activities are based on the household labor and agricultural labor modules in the post-planting and post-harvest surveys where respondents indicate which members are engaged in different activities. We code each member as engaged in a given activity for a survey round if they were engaged in it during either the post-planting or post-harvest round, then take the sum of members in each activity for the household. Hours of work over the last 7 days by household members are aggregated from the same survey modules.

Section **??** describes how the earnings and value of production variables are constructed. Farm production includes crop and livestock activities. Non-farm earnings include household non-farm enterprise, wage employment, and 'other activities' including income from pensions, remittances, investments, transfers, etc. Households not active in a given activity are assigned an income of zero for that activity. We drop households with zero total earnings and production value in 2010-11 from the analysis as these likely represent missing data rather than true zeros. Households with zero total earnings and production value are more prevalent in 2010-11 (11.1% of the sample) than in subsequent rounds (3.8%) and do report being engaged in farm production (43%), NFE (51%), and wage employment (23%), indicating challenges with collecting earnings or production data in the first implementation of the survey. Results are qualitatively similar but with slightly smaller coefficient magnitudes if these households are included. The household earnings Herfindahl-Hirschman-Index (HHI) is the sum of squared shares of earnings (without subtracting expenses) from farm production, NFE, wage work, and other sources. Values closer to one indicate a greater concentration of income sources.

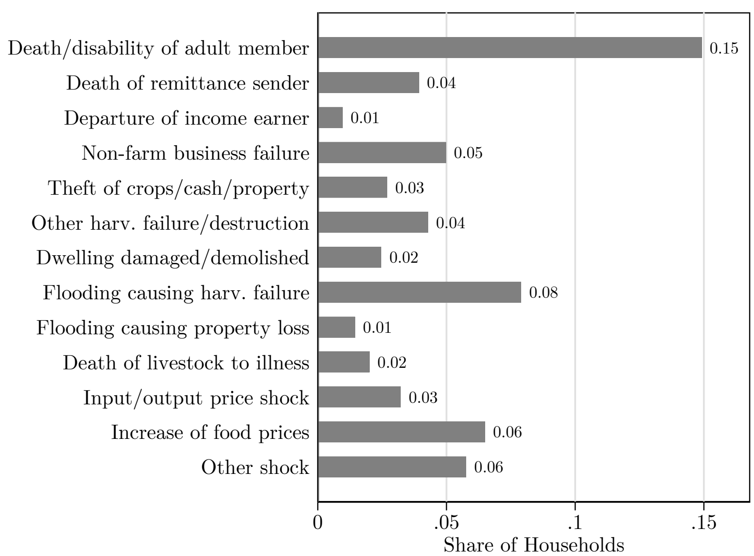
All crop production outcomes refer to the most recent growing season and are aggregated across plots. Outcomes for households not active in crop production in a given round are assigned values of 0, except for the crop area HHI and proportion of crop value sold which are undefined for households with 0 area planted and the crop yield and price values which are undefined for households not cultivating particular crops. The number of crops cultivated is across all household plots. The crop area HHI is the sum of squared shares of total planted area allocated to different crops. Values closer to 1 indicate less diversification across crops. Total hired/exchange and family labor days and family labor hours are summed across all crop production activities reported in the post-planting and post-harvest surveys. Crop production costs are the sum of expenditures on purchased inputs such as seeds, fertilizer, labor, land, etc. Crop sales are the sum across crops of quantity sold times sales price. Crop production value is the sum across crops of crop sales plus the value of unsold crops, measured as quantity harvested but not sold times either the median household sale price, the median sales price for that crop in the community if the household did not sell but at least three community households did, or the median price at the lowest administrative level with at least three sales observations. The proportion of value of crop production sold is the ratio of the two above variables.

Physical crop yields are calculated as the total quantity harvested divided by the area planted. Crop sales prices are defined in the calculation of crop production value. For cereals, we calculated mean yields by taking total cereal quantity harvested over total cereal area planted and calculated quantity-weighted mean prices across different cereals produced. We calculate a household crop yield index and crop price index across the eight most commonly cultivated crops in the sample: maize, rice, sorghum, millet, cowpea, groundnut, yam, and cassava. We first normalize yield and sales price within survey round for each crop, and then calculate area planted-weighted means of normalized yields and prices across crops produced. We then normalize these indices within survey round.

Differences across survey flood measures

Several survey questions ask about flood exposure, each of which captures specific ways in which floods can affect households. In the post-planting agricultural survey, households can list flooding as the main cause of loss of stored crops since the beginning of the new year for each cultivated crop. Such losses are reported by 13% of households for 2012 but flood-related losses are only reported by 1% In the post-harvest agricultural survey, households can list flooding as a reason for not harvesting a particular crop. Failure to harvest is reported by 24% of households, but floods are only listed as a reason by 3%. In both the post-planting and post-harvest household surveys, households can list flooding as a cause of household food insecurity in the past 12 months. Food insecurity is reported by 24% of households at post-planting and by 20% at post-harvest in the 2012-13 survey, and in both periods 3% list floods as a cause of food insecurity. Finally, in the post-harvest household survey shocks module, households are asked about various shocks they have been affected by over the past 5 years, the years in which they occurred, and the consequences of the shocks. Two shocks relate to floods: flooding that caused harvest failure is reported by 7% of households in 2012 and loss of property due to flood is reported by 1%. Floods are the second most commonly reported shock for 2012 (Figure A6).

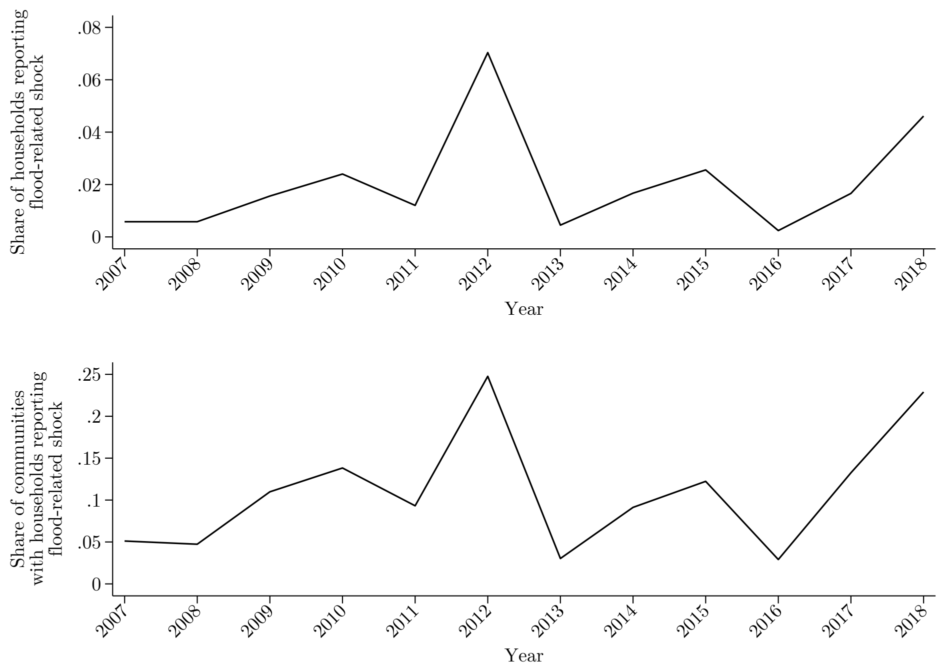
Figure A6: Prevalence of shocks reported by households over the 12 months before the 2013 post-harvest survey round



Note: Data are from the 2013 post-harvest household survey shocks module.

We can use the fact that household shocks are reported for multiple years to show how the prevalence of flood shocks was much greater in 2012 than in any other year from 2007-2018 (Figure A7). The fact that the share of households reporting flood shocks is much lower than the share of communities where such shocks are reported highlights how not every household is affected in communities where flooding occurs.

Figure A7: Survey flood shock reports by year

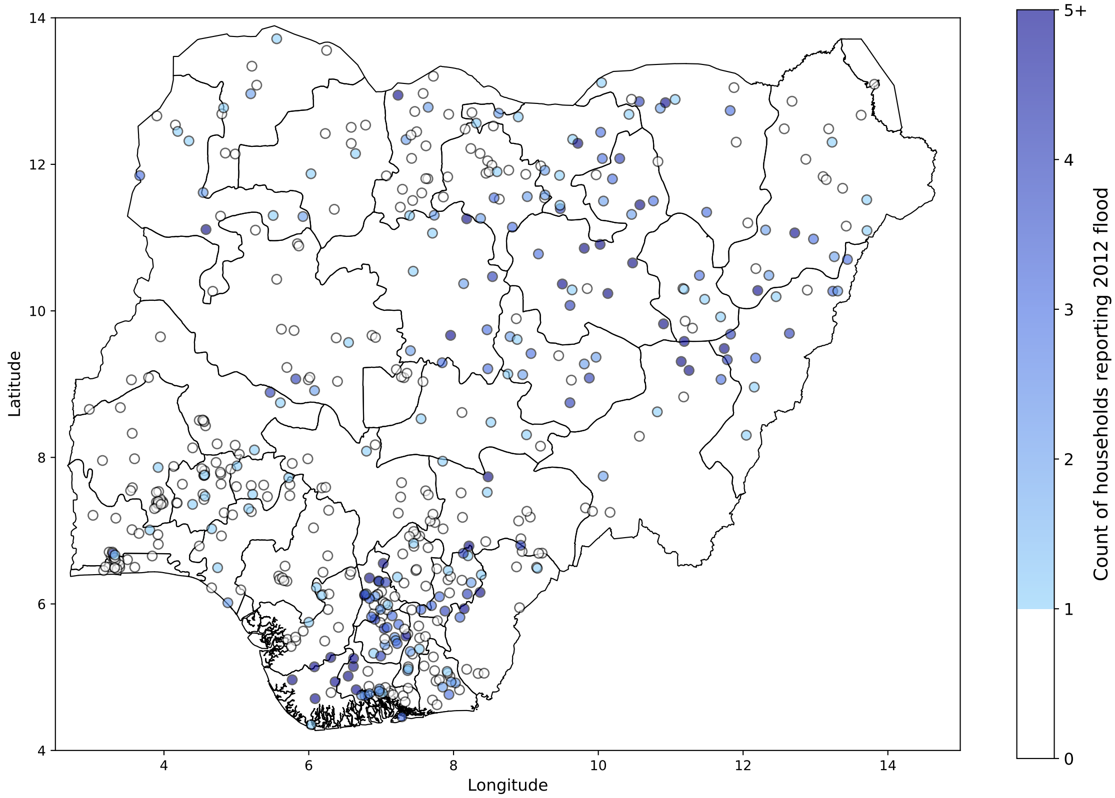


Note: Annual flood reports are from the Nigeria GHSP. In each survey round, households are asked to recall whether they experienced specific economic shocks over the last few years, including flooding that caused harvest failure and loss of property due to flood. These data were collected in the post-harvest survey waves in 2009, 2013, 2016, and 2019. The top panel shows the share of households reporting any flood shock by year. The bottom panel shows the share of communities with any household flood shock report by year . This is a more restrictive measure of flood exposure than the main survey-based measured we use in this paper which incorporates reports of floods in the survey's food security and crop production modules and in the community informant survey, but there is no recall element for the other household survey questions.

Across all flood questions, 12.4% of households report being affected by floods in some way in 2012. **??** visualizes the count of households reporting floods across communities, further demonstrating how in most communities where floods are reported, only a subset of the ten sample households report being affected. This result is supported by the community questionnaire, which asks informants to report important events that made people worse off in the community, including floods, and the year in which they occurred. Floods are reported in 20% of communities in 2012, and on average these are reported to have affected 48% of community households (the median is

50%).

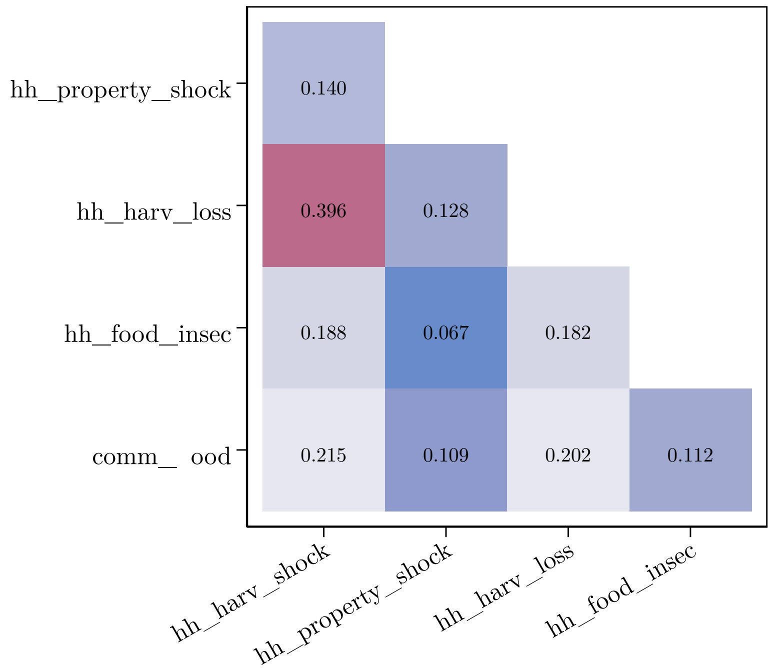
Figure A8: Counts of households reporting flood exposure in 2012 by community



Note: Nigeria GHSP household flood reports are aggregated across questions from the household shocks, food security, and crop production modules. For each community, we calculate the number of households reporting being affected by floods in 2012 in one of these modules. White circles indicating communities with no household flood reports. Ten households are sampled for each survey community.

The main survey flood definition is based on the union of reports of floods from all household survey questions and from the community survey. Figure A9 shows a heatmap of correlations between these measures. We aggregate the agricultural survey questions as 'hh_harv_loss' and the food insecurity questions as 'hh_food_insec', in both cases coded as 1 if the household reports being affected by a flood in either the post-planting or post-harvest wave. We compare these against the two flood questions from the household shock module ('hh_harv_shock' and 'hh_property_shock') and against reports of floods that negatively affected community members by informants in the community survey ('comm_flood'). While the two harvest-related measures are strongly correlated, correlations between the other measures are weaker. This emphasizes that each measure is capturing different ways in which floods may affect households, and that they primarily affect different households rather than the same household being affected in multiple ways. The figure also indicates that household flooding reported in the shocks module – a common way survey-based flood measures are constructed – may miss some floods that did not cause the specific types of losses or damages that are asked about.

Figure A9: Correlations between survey measures of household-level 2012 flood exposure reports



Note: This heatmap shows pairwise correlations between different measures of flood exposure in 2012 at the level of households in the Nigeria GHSP. 'hh_harv_shock' indicates a report of a flood that caused harvest failure in the household shocks module. 'hh_property_shock' indicates a report of a flood that caused property loss in the household shocks module. 'hh_harv_loss' indicates a report of loss of stored crops or failure to harvest in the post-planting and post-harvest household crop production modules. 'hh_food_insec' indicates a report of floods as a cause of household food insecurity in either the post-planting or post-harvest food security modules. 'comm_flood' indicates that informants in the community survey report that floods negatively affected community members.

How do these household flood reports compare to what is reported in the GHSP community surveys? Twenty percent of households reside in communities where flooding was reported in 2012 in the community survey. The survey and community measures are correlated: in communities with a community survey flood report, 25% of households report a flood, compared to 7% in communities with no community flood report. But this mismatch leads to 41% of households that report being affected by floods in 2012 residing in communities with no community survey flood report.

Households reporting floods not captured in the community survey may be driven by lack of complete information by the group of community survey informants, or by error in the household surveys. Even in communities with at least two (of ten sample households) separate households reports of flood in the shocks module, some community surveys do not report flooding. Community flood reports are more strongly correlated with household reports of harvest losses than with measures of property loss or food insecurity. This suggests differences in the types of damages that are more salient to community informants. On the other hand, there are 41 communities with a community flood report but no survey flood report. A potential reason for this is floods affecting community households other than those included in the GHSP sample, or affecting households in ways not captured in the household survey questions.

D Flood detection with MODIS satellite imagery

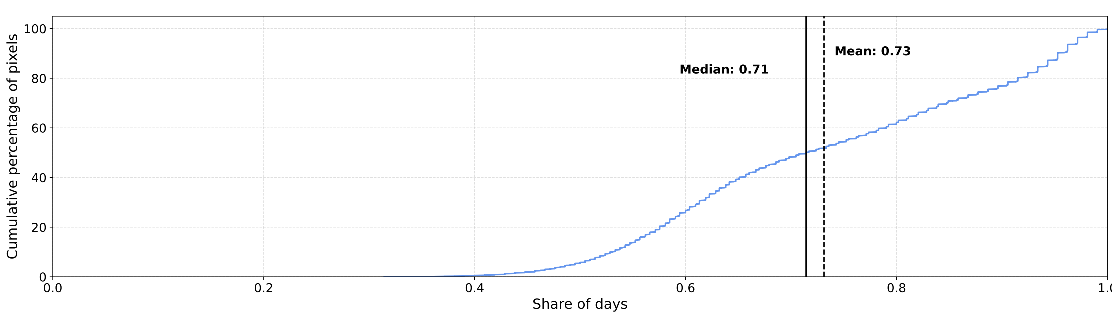
This appendix illustrates reasons why remote detection of flooding using MODIS satellite imagery, as used in this paper via the NASA Global Flood Product, may not align with floods reported in surveys. First, optical satellite imagery is constrained by cloud cover which can limit the ability to observe changes in surface water. Second, MODIS has relatively coarse spatial resolution, limiting its ability to detect narrow or localized water features. Third, the satellite-based community indicator is constructed using a binary intersection rule: a community is classified as flooded whenever the any 2012 flooding is including in the NASA Global Flood Product within a 5 km buffer. Floods detected within this buffer may not affect survey households for a variety of reasons.

D.1 Cloud cover and observability

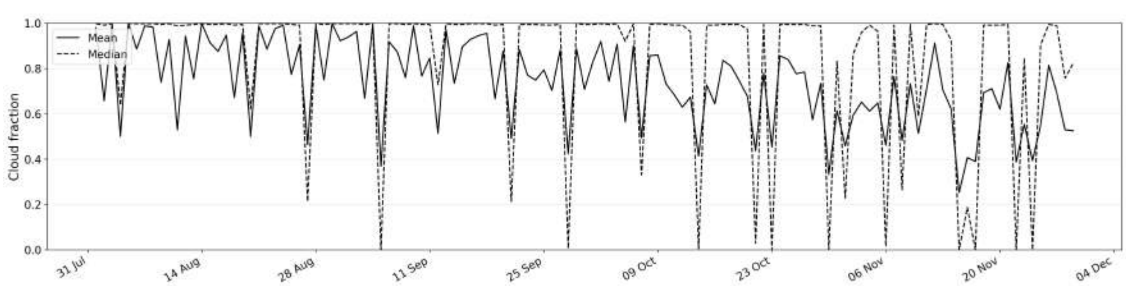
Optical satellite imagery cannot detect floods when the Earth's surface is covered by clouds. Figure A10 documents cloud conditions during the 2012 flood season in Nigeria, between August and November. Panel A shows that the median MODIS pixel (250 m) in Nigeria was cloudy on roughly 71% of days, and the mean was approximately 73%. This indicates a broad national pattern of frequent cloud cover during the months in which flooding was most likely to occur.

Figure A10: Cloud cover in MODIS satellite imagery in Nigeria, August-November 2012

(a) Distribution of share of cloudy days across all Nigeria pixels



(b) Daily mean and median cloud fraction within 5 km buffers around GHSP communities



Note: Data are from the MOD09GA collection on Google Earth Engine.

Panel B focuses on the empirical setting of the paper by plotting the daily mean and median cloud fraction within 5 km buffers around GHSP communities. Cloud cover remains high for much of the flood season, with means around 80-90% from August-September and 50-80% from October-November. The median is often close to one, implying that on most dates at least half of the surveyed communities were almost entirely cloud-covered. Although the cloud fraction occasionally drops sharply, these clearer observation windows are brief and irregular.

Taken together, these patterns indicate that cloud cover during the period when most floods occurred in 2012 is a major constraint to remote sensing of flooding using MODIS imagery. The constraint implies that the absence of flooding in the NASA database cannot be interpreted as evidence that no flooding occurred on the ground, and likely explains a large share of the failure of the database to identify flooding in communities with survey-reported floods.

D.2 Spatial resolution and water detection

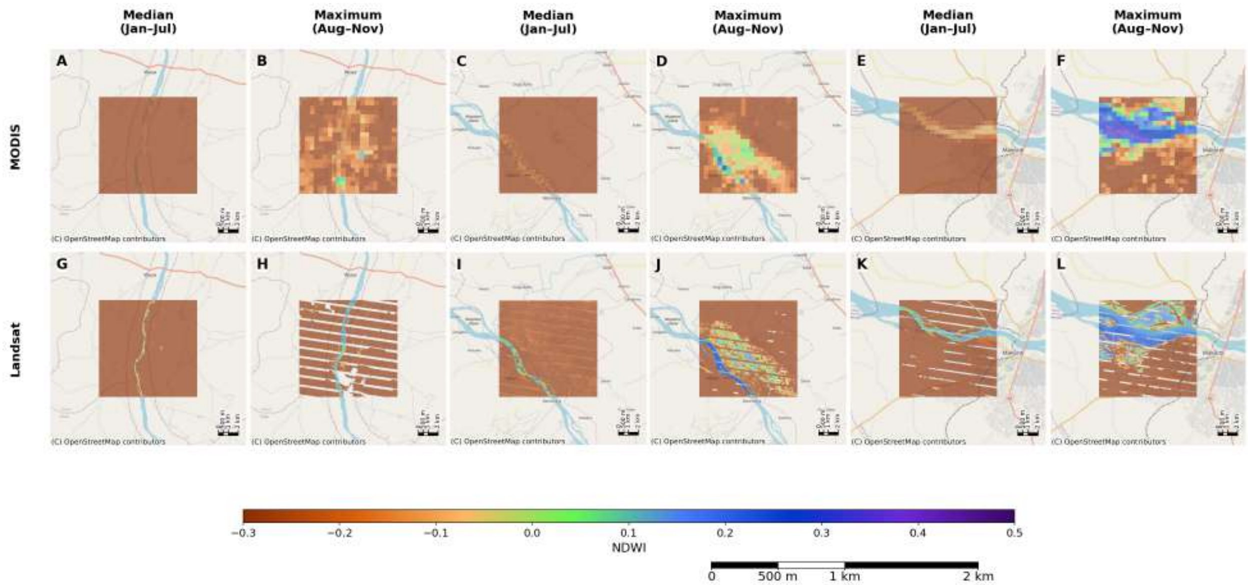
Another limitation of MODIS imagery is its 250 m resolution. Detection of surface water is based on reflection of different bandwidths of light, formalized in the Normalized Difference Water Index (NDWI). Pixels of mixed water and other surfaces will have lower NDWI values than pixels with a large share of water cover, and the water cover is therefore less likely to be identified. We illustrate how the MODIS data resolution limits its ability to detect water in comparison with Landsat 7 satellite imagery, which has a 30 m resolution but is collected less frequently.

For both sources, we calculate the McFeeters NDWI (McFeeters, 1996), which is designed to enhance open surface water, rather than the Gao NDWI (Gao, 1996), which is more closely related to vegetation moisture. NDWI values above 0 or 0.1 are typically taken to likely represent surface water, with higher values indicating greater certainty. During the non-flood period (January–July 2012), we use median NDWI to represent typical surface-water conditions. During the flood period (August–November 2012), we use maximum NDWI to give the imagery the best possible chance to reveal changes in surface water.

Figure A11 shows three cases illustrating how the ability of 250 m resolution MODIS imagery to detect water depends on the spatial scale of the underlying feature, depicted using data from OpenStreetMap. In the first case, MODIS detects no meaningful variation in NDWI above the narrow river in the non-flood season (Panel A), and the signal remains unclear in the flood season

(B). In the second case, with a slightly larger river, MODIS identifies a different NDWI signal during the non-flood period but it is not above typical thresholds for classifying water (C). It does identify some clear river water cells in the flood season, as well as some cells with potential flooding (D). In the third case, with a river around 500 m in width at many points even in the dry season, MODIS imagery better identifies cells with higher NDWI where the river lies during the non-flood season (E), but these NDWI values are not high enough to clearly identify water though identification is stronger during the flood season (F).

Figure A11: Progression in water detection across MODIS and Landsat imagery.



Note: Data are from the 250 m MODIS MOD09GA (top) and 30 m Landsat LE07/C02/T1_L2 (bottom) collections on Google Earth Engine. All data are from 2012. We present three cases with rivers of different widths to illustrate how the ability of MODIS to detect surface water depends on the spatial scale of the underlying feature. For each case, we visualize median NDWI during the January-July non-flood period and maximum NDWI during the August-November flood period for both MODIS and Landsat. NDWI values above 0 are more likely to indicate the presence of surface water.

In all cases, the 30 m resolution Landsat 7 imagery (bottom row) clearly identifies water cells above the rivers in the dry non-flood season, as well as expansions and overflows of these rivers during the flood season. In all cases, the higher resolution Landsat imagery provides sharper and more precise information on river channel structure and likely flood overflows.

The fact that MODIS imagery fails to detect even large rivers during certain periods clearly shows its limitations for detecting floods. Floods must spread over a large share of the 250 m pixel area to be detectable using NDWI-based algorithms. Floods that spread unevenly over space – likely in uneven terrain – are unlikely to be detected. MODIS imagery is therefore likely capture only a subset of more severe floods or floods in particular geographic conditions, which can further help explain the large number of GHSP communities with 2012 floods not identified in the NASA database.

D.3 Illustrating survey vs. satellite flood detection

To illustrate how survey-based and satellite-based flood measures can agree or disagree at the community level, we present visual evidence from selected community examples. These examples are not intended to validate one source against the other mechanically. Rather, they show how the spatial distribution of water, uncertainty about the true location of the community within the buffer, and the construction of the MODIS-based indicator help interpret the relationship between the two measures.

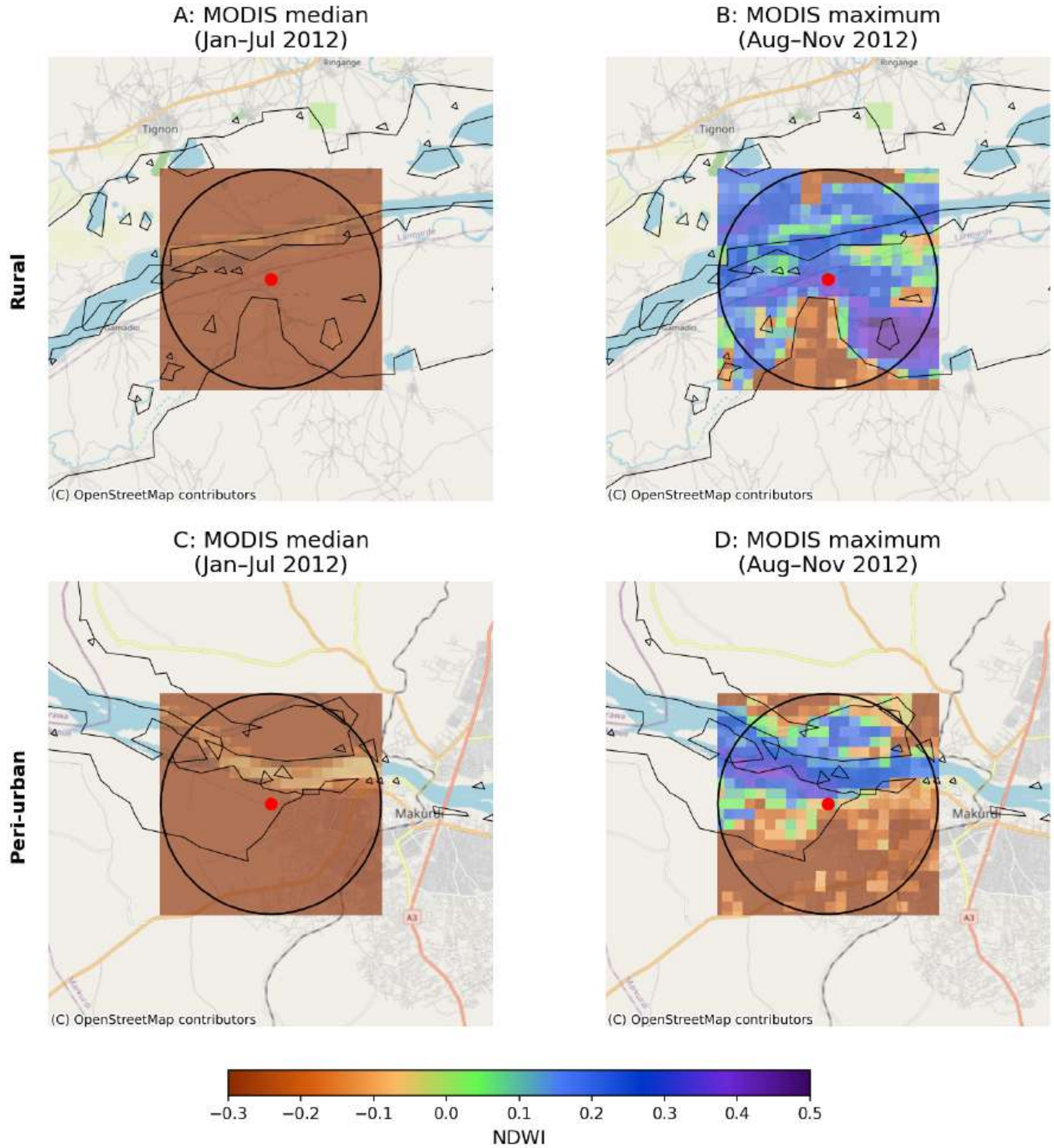
Figure A12 presents two cases of agreement between the survey and satellite flood measures we use, considering rural and peri-urban communities near rivers. In both cases, we find much higher NDWI values in MODIS data during the flooding period than during the non-flooding period, with clear alignment with the MODIS-based flood detection boundaries. The figures appear to show clear cases of riverine flooding in large parts of the 5 km buffer around the communities.

Figure A13 presents examples of disagreement between the two flood measures. Panel a) shows two communities where flooding is detected within 5 km of the community centroid by the NASA GLobal Flood Product, but is not reported in any household or community surveys.[1] In the top row, changes in NDWI during the flooding season suggest a broader increase in local surface water in what appears to be a hydrologically complex setting characterized by marshes, so the satellite flood identification may reflect real water expansion that was not experienced as a flood. In the bottom row, visible change in NDWI within the buffer is much weaker, yet the community is still classified as flooded because the NASA flood layer intersects the 5 km buffer. The mismatch may represent either a limitation of the buffer based matching process for the NASA flood layer – required due to uncertainty about exact community locations – or a limitation or error in the NASA flood layer itself.

Panel b) illustrates the case of a community with multiple survey flood reports but no flood detection within 5 km in the NASA database. Changes in NDWI captured by MODIS suggest

[1] In 17 of 73 communities with satellite-only flooding, there is no flood according to the main survey-based measure requiring at least 2 household flood reports, but there is at least one household or community survey flood report.

Figure A12: Examples of survey-satellite agreement in rural and urban settings

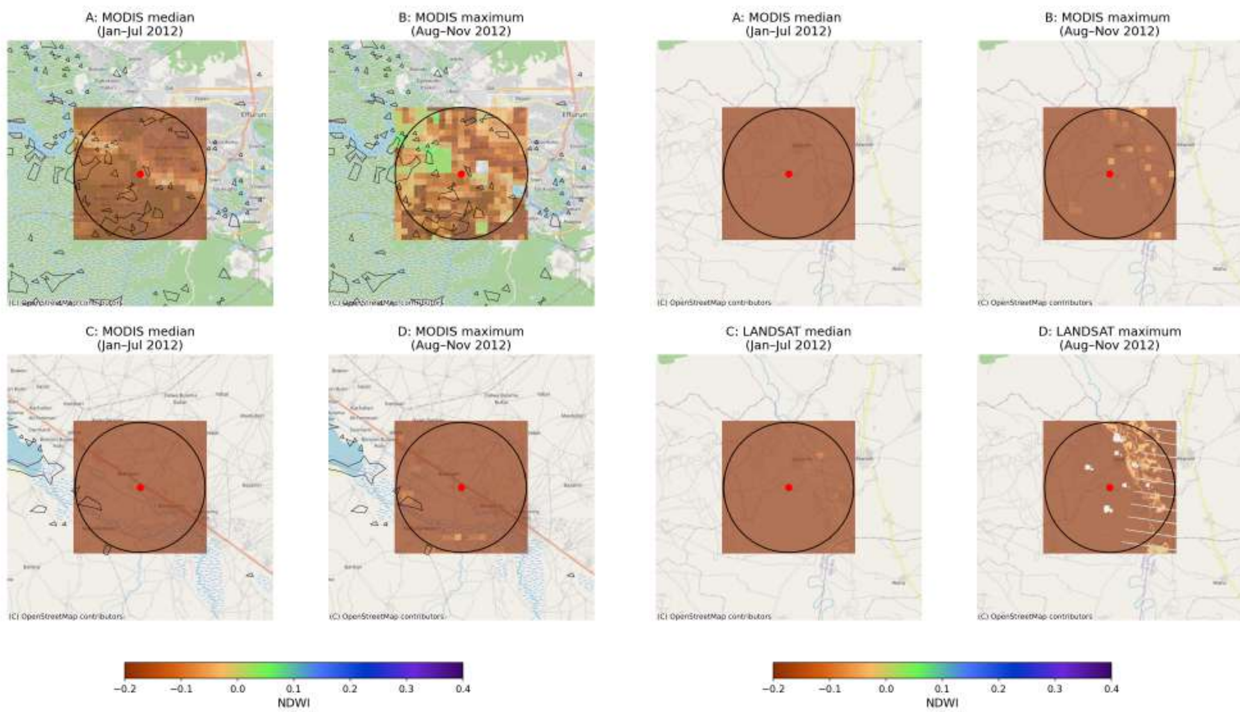


Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. The left panels show median MODIS NDWI in the non-flood period (January–July 2012), while the right panels show maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA Global Flood Product layer used for the satellite-based flood measure.

only weak differences during the flood season, but Landsat data suggests a broader pattern of localized riverbed expansion, some of which may have been experienced as flooding. This is the type of setting in which spatial resolution matters most: localized flooding may be meaningful for households while remaining only weakly visible in MODIS.

Figure A13: Examples of survey-satellite disagreement

(a) Satellite-only cases (b) Survey-only case



Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. For each pair of images, the left panel shows median MODIS NDWI in the non-flood period (January–July 2012), while the right panel shows maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA Global Flood Product layer used for the satellite-based flood measure. Panel A presents two communities classified as flooded by the MODIS-based indicator but with no survey flood reports. Panel B presents a community with at least two survey flood reports no MODIS-based flooding detected within 5 km. For comparison, we also show changes in NDWI using higher-resolution Landsat which suggests a clearer pattern of riverbed expansion.

References

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